

# Ethan Levy

📍 Los Angeles, CA | ✉ ethan9000@outlook.com | 📞 +1 (301) 655-1775 | 🌐 levypedia.com

## Education

**University of Southern California**  
*M.S. in Electrical Engineering, GPA: 3.85*

Los Angeles, CA  
*Aug 2024 – Dec 2025*

**University of Maryland**  
*B.S. in Physics (Honors), Minor: Sustainability, GPA: 3.59*  
Awards: Angelo Bardasis Memorial Scholar (4x), Dean's List (2021, 2023)

College Park, MD  
*Sep 2020 – May 2024*

## Technical Skills

**Simulation:** COMSOL Multiphysics, Lumerical FDTD, SolidWorks  
**Programming:** Python, MATLAB, Mathematica, HTML/CSS, Github, LaTeX  
**Specialization:** Optics & Photonics, Light-Matter Interaction, Quantum Optics, Numerical Simulation

## Research Experience

**Research Assistant, Nanostructure Materials and Devices Laboratory** Mar 2025 – Oct 2025  
*University of Southern California Los Angeles, CA*

- Conducted photoluminescence spectroscopy and time-resolved measurements on spatially ordered arrays of single-photon emitting quantum dots at above-gap excitation to assess single-photon emission quality
- Performed  $g^{(2)}$  second-order correlation measurements to calculate single-photon statistics and characterize quantum emitter performance in monolayer transition metal dichalcogenide quantum dots

**Research Assistant, Joint Quantum Institute** Jun 2023 – Feb 2024  
*University of Maryland College Park, MD*

- Designed and executed FDTD simulations of a topological photonic beam splitter using Lumerical
- Performed parameter sweeps to optimize waveguide confinement and transmission efficiency, analyzing mode propagation and coupling characteristics

**Research Assistant, Compact Muon Solenoid (CMS) Collaboration** Dec 2023 – May 2024  
*University of Maryland College Park, MD*

- Utilized DD4hep and Geant4 simulation frameworks to model dual-readout calorimetry, analyzing Cerenkov and scintillation light detection in hadronic fibers
- Presented poster “DD4hep as a Tool for Studying Dual-Readout Calorimetry” at Second Annual U.S. Future Circular Collider Workshop, MIT (March 2024)

**Research Assistant, Supersymmetry Theory Group** Jun 2022 – Nov 2023  
*University of Maryland College Park, MD*

- Conducted theoretical research in supersymmetry under Prof. S. James Gates Jr. through SSTPRS REU program
- Applied differential geometry, tensor algebra, gauge theory, and Clifford/Lie algebras to investigate supermultiplet properties, co-authoring publication on Adinkra representations

## Publications

Cianciara, A.J., Gates Jr., S.J., **Levy, E.T.**, Razzaz, T.O., & Richardson, J. (2023). “Unfolded Adinkra Properties of Supermultiplets (I).” *arXiv preprint arXiv:2311.06842*.

## Teaching & Leadership

**Teaching Assistant** 2022 – 2023  
*University of Maryland College Park, MD*

- PHYS132: Led discussion and lab sessions in Electricity & Magnetism and Waves for life sciences students (Spring 2023)
- PHYS105: Facilitated weekly discussions on energy production, storage, and thermodynamics in climate change context (Fall 2022)

**Eagle Scout** 2012 – 2019  
*Boy Scouts of America*